

A Structural Reduction for the Compact Shell Proxy

Round 1 analytic note

Abstract

This note isolates a possible purely analytic replacement for the compact interval certificate in the spherical-shell argument. The weighted proxy is first written as an exact layer-cake sum. Its deficit from the action integral is then decomposed into a radial shifted-quadrature deficit and an angular rounding defect. A Stieltjes calculation retains two nonnegative remainders that are discarded by the existing optical estimate. One of those remainders admits a smooth explicit lower bound. The result is a single concrete inequality, displayed in (37), whose proof would close the actual compact residual without numerical subdivision. That final inequality is not proved here.

1 Action and the compact target

For $a > 0$ and $z \geq 0$, put

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z < a, \\ 0, & z \geq a. \end{cases} \quad (1)$$

Fix $0 < \rho < 1$ and $K > 0$, and define

$$A(z) = A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z), \quad T := A(0) = \frac{(1-\rho)K}{\pi}. \quad (2)$$

The strict positive-integer count is

$$[x]_{<} := \#\{n \in \mathbb{N} : n < x\}.$$

The compact proxy and its continuous payment are

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2 < K \\ \ell \in \mathbb{N}_0}} 2\nu \left[A(\nu) + \frac{1}{4} \right]_{<}, \quad (3)$$

$$W(\rho, K) := \int_0^K 2z A(z) dz = \frac{2(1-\rho^3)K^3}{9\pi}. \quad (4)$$

The actual compact residual is contained in

$$\mathcal{R}_c := \left\{ (\rho, K) : \frac{1}{1+2\pi} \leq \rho \leq \frac{39}{50}, \quad k_{11}(\rho) \leq K \leq 200 \right\}, \quad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}. \quad (5)$$

Thus it is enough to prove $P < W$ on $\mathcal{R}_c$.

For $0 < z < K$, direct differentiation gives

$$-A'(z) = \frac{1}{\pi} \begin{cases} \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}, & 0 < z < \rho K, \\ \arccos \frac{z}{K}, & \rho K \leq z < K. \end{cases} \quad (6)$$

In particular, A is strictly decreasing from T to 0. Let

$$R : [0, T] \longrightarrow [0, K]$$

be its decreasing inverse, and set

$$F(t) := R(t)^2, \quad g(t) := -F'(t), \quad \tau := A(\rho K) = KG_1(\rho). \quad (7)$$

The derivative in (6) increases in magnitude on $(0, \rho K)$ and decreases in magnitude on $(\rho K, K)$. The inverse curvature has the related, but slightly stronger, property

$$g \text{ decreases on } (0, \tau), \quad g \text{ increases on } (\tau, T). \quad (8)$$

Indeed, on the outer branch, if $z = R(t)$ then

$$g(t) = \frac{2\pi z}{\arccos(z/K)};$$

the quotient increases with z , while z decreases with t . On the inner branch, put

$$\delta(z) := \arccos \frac{z}{K} - \arccos \frac{z}{\rho K}.$$

Differentiation with respect to the intermediate ratio gives

$$\frac{\delta(z)}{z} = \int_{\rho}^1 \frac{ds}{s\sqrt{s^2 K^2 - z^2}}.$$

This integral increases with z . Thus $z/\delta(z)$ decreases with z , and $g(t) = 2\pi z/\delta(z)$ increases as t increases and z decreases. The endpoint and interface data are

$$F(\tau) = \rho^2 K^2, \quad g(\tau) = \frac{2\pi \rho K}{\arccos \rho}, \quad g(T) = \frac{2\pi \rho K}{1 - \rho}. \quad (9)$$

Here and below $g(T)$ is the one-sided limit. Near $t = 0$, one has $g(t) = O(t^{-1/3})$, so all improper Stieltjes boundary terms used below vanish.

2 Exact layer cake and angular rounding

Let

$$x_n := n - \frac{1}{4}, \quad M(r) := \# \left\{ \ell \in \mathbb{N}_0 : \ell + \frac{1}{2} < r \right\} = \max \left\{ 0, \left\lceil r - \frac{1}{2} \right\rceil \right\}. \quad (10)$$

Lemma 1 (Exact layer cake). *For every $0 < \rho < 1$ and $K > 0$,*

$$P(\rho, K) = \sum_{\substack{n \geq 1 \\ x_n < T}} M(R(x_n))^2, \quad W(\rho, K) = \int_0^T F(t) dt. \quad (11)$$

Both identities retain the strict convention at every radial and angular wall.

Proof. For a fixed half-integer ν ,

$$\left[A(\nu) + \frac{1}{4} \right]_{<} = \#\{n \geq 1 : x_n < A(\nu)\}.$$

Since A and R are decreasing, $x_n < A(\nu)$ is equivalent to $\nu < R(x_n)$. Interchanging the two finite sums and using

$$\sum_{\ell=0}^{m-1} (2\ell + 1) = m^2$$

proves the first identity. At $R(x_n) = m + 1/2$, the strict angular count is m , exactly as specified by M .

For the second identity, the area formula for a decreasing inverse gives

$$\int_0^T R(t)^2 dt = \int_0^K 2zA(z) dz.$$

The last integral is (4). □

For every active n , abbreviate

$$r_n := R(x_n), \quad m_n := M(r_n), \quad \alpha_n := r_n - m_n + \frac{1}{2}. \quad (12)$$

The strict convention implies

$$m_n - \frac{1}{2} < r_n \leq m_n + \frac{1}{2}, \quad 0 < \alpha_n \leq 1. \quad (13)$$

This includes $m_n = 0$. Define the angular rounding defect

$$\mathcal{E}_{\text{ang}}(\rho, K) := \sum_{x_n < T} (m_n^2 - r_n^2). \quad (14)$$

The defect of a single radial layer is exactly

$$m_n^2 - r_n^2 = (1 - 2\alpha_n)r_n + \left(\alpha_n - \frac{1}{2} \right)^2. \quad (15)$$

Unlike the one-sided estimate $m_n^2 - r_n^2 < r_n + 1/4$, formula (15) retains the cancellation when $\alpha_n > 1/2$.

Combining (11) and (14) gives the first exact two-term decomposition:

$$W - P = \mathcal{D}_{\text{rad}} - \mathcal{E}_{\text{ang}}, \quad \mathcal{D}_{\text{rad}} := \int_0^T F(t) dt - \sum_{x_n < T} F(x_n). \quad (16)$$

3 The radial deficit with retained positive remainders

Define the shifted counting function and its two primitives by

$$C(t) := \#\{n \geq 1 : x_n < t\}, \quad w(t) := t - C(t), \quad \mathfrak{W}(t) := \int_0^t w(s) ds, \quad \Psi(t) := \mathfrak{W}(t) - \frac{t}{4}. \quad (17)$$

On $0 \leq t \leq 3/4$,

$$\Psi(t) = \frac{1}{2} \left(t - \frac{1}{4} \right)^2 - \frac{1}{32}. \quad (18)$$

For $t = x_n + u$, $0 \leq u \leq 1$,

$$\Psi(t) = \frac{1}{2} \left(u - \frac{1}{2} \right)^2 - \frac{1}{32}. \quad (19)$$

Consequently

$$\mathfrak{W}(t) \geq 0, \quad -\frac{1}{32} \leq \Psi(t) \leq \frac{3}{32} \quad (t \geq 0). \quad (20)$$

Proposition 2 (Exact retained-remainder identity). *Put*

$$\mathcal{R}_{\text{dec}} := - \int_{(0,\tau)} \mathfrak{W}(t) dg(t), \quad (21)$$

$$\mathcal{R}_{\text{osc}} := \left(\Psi(T) + \frac{1}{32} \right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi(t) \right) dg(t). \quad (22)$$

Then both remainders are nonnegative, and

$$\begin{aligned} \mathcal{D}_{\text{rad}} &= \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau) \left(\frac{\tau}{4} + \frac{3}{32} \right) \\ &\quad + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}}. \end{aligned} \quad (23)$$

Therefore

$$W - P = \mathcal{B}(\rho, K) + \mathcal{R}_{\text{dec}} + \mathcal{R}_{\text{osc}} - \mathcal{E}_{\text{ang}}, \quad (24)$$

where

$$\mathcal{B}(\rho, K) := \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{KG_1(\rho)}{4} + \frac{3}{32} \right). \quad (25)$$

Proof. Distributionally, $dw = dt - dC$. Stieltjes integration by parts, using $F(T) = 0$ and $w(0) = 0$, gives

$$\mathcal{D}_{\text{rad}} = \int_0^T w(t)g(t) dt. \quad (26)$$

The improper boundary term at 0 vanishes because $\mathfrak{W}(t) = t^2/2$ there and $g(t) = O(t^{-1/3})$.

On $(0, \tau)$, integrate with $d\mathfrak{W} = w dt$:

$$\int_0^\tau wg = \mathfrak{W}(\tau)g(\tau) - \int_{(0,\tau)} \mathfrak{W} dg. \quad (27)$$

On (τ, T) , use $w = 1/4 + \Psi'$ and $\int_\tau^T g = F(\tau)$:

$$\int_\tau^T wg = \frac{F(\tau)}{4} + \Psi(T)g(T) - \Psi(\tau)g(\tau) - \int_{(\tau,T)} \Psi dg. \quad (28)$$

Since $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, adding (27) and (28) cancels the uncontrolled interface value $\Psi(\tau)$.

Now add and subtract the extremal sawtooth values in (20). The exact identity

$$\begin{aligned} &\Psi(T)g(T) - \int_{(\tau,T)} \Psi dg \\ &= -\frac{g(T)}{8} + \frac{3g(\tau)}{32} + \left(\Psi(T) + \frac{1}{32} \right) g(T) + \int_{(\tau,T)} \left(\frac{3}{32} - \Psi \right) dg \end{aligned}$$

gives (23). By (8), $-dg$ is a positive measure on $(0, \tau)$ and dg is a positive measure on (τ, T) . Equations (20), (21), and (22) therefore prove nonnegativity of both remainders. Finally substitute (9) and $\tau = KG_1(\rho)$, and then use (16). $\square$

4 A smooth lower bound for the decreasing-branch remainder

The term $\mathcal{R}_{\text{dec}}$ is quantitatively important on the compact residual; discarding it loses too much. It has the following elementary smooth minorant.

Lemma 3 (A global primitive minorant). *For $t \geq 0$, define*

$$L(t) := \frac{t^2}{2(1+2t)}, \quad L'(t) = \frac{t(1+t)}{(1+2t)^2}. \quad (29)$$

Then

$$0 \leq L(t) \leq \mathfrak{W}(t) \quad (t \geq 0). \quad (30)$$

Proof. For $0 \leq t \leq 3/4$, one has $\mathfrak{W}(t) = t^2/2 \geq L(t)$. For $t \geq 3/4$, (20) gives

$$\mathfrak{W}(t) \geq \frac{t}{4} - \frac{1}{32}.$$

Moreover,

$$\frac{t}{4} - \frac{1}{32} - \frac{t^2}{2(1+2t)} = \frac{6t-1}{32(1+2t)} \geq 0.$$

This proves (30). $\square$

Define

$$\mathcal{R}_*(\rho, K) := - \int_{(0, \tau)} L(t) dg(t). \quad (31)$$

Because $-dg$ is positive on the decreasing branch,

$$0 \leq \mathcal{R}_* \leq \mathcal{R}_{\text{dec}}. \quad (32)$$

Integration by parts is legitimate at 0 because $L(t)g(t) = O(t^{5/3})$. Hence

$$\mathcal{R}_* = -L(\tau)g(\tau) + \int_0^\tau L'(t)g(t) dt. \quad (33)$$

On the outer branch $t = A(z)$, one has $g(t) dt = -2z dz$. Therefore

$$\mathcal{R}_* = -\frac{\tau^2}{2(1+2\tau)}g(\tau) + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz. \quad (34)$$

This expression is smooth in (ρ, K) on the compact residual and has no floor or wall discontinuity.

Corollary 4 (A concrete sufficient inequality). *Define*

$$\begin{aligned} \mathcal{H}(\rho, K) := & \frac{\rho^2 K^2}{4} - \frac{\pi \rho K}{4(1-\rho)} \\ & + \frac{2\pi \rho K}{\arccos \rho} \left(\frac{\tau}{4(1+2\tau)} + \frac{3}{32} \right) \\ & + \int_{\rho K}^K 2z \frac{A(z)(1+A(z))}{(1+2A(z))^2} dz, \quad \tau = KG_1(\rho). \end{aligned} \quad (35)$$

If

$$\mathcal{E}_{\text{ang}}(\rho, K) < \mathcal{H}(\rho, K), \quad (36)$$

then $P(\rho, K) < W(\rho, K)$.

Proof. By (25), (34), and

$$\frac{\tau}{4} - \frac{\tau^2}{2(1+2\tau)} = \frac{\tau}{4(1+2\tau)},$$

one has $\mathcal{H} = \mathcal{B} + \mathcal{R}_*$. Now use (24), (32), and $\mathcal{R}_{\text{osc}} \geq 0$. $\square$

5 The first unresolved inequality

The preceding calculation reduces the analytic replacement project to the following statement:

$$\boxed{\sum_{\substack{n \geq 1 \\ n-1/4 < T}} (M(R(n-1/4))^2 - R(n-1/4)^2) < \mathcal{H}(\rho, K)} \quad ((\rho, K) \in \mathcal{R}_c). \quad (37)$$

This is the first unresolved inequality in the present route. Everything to its right is the smooth action expression (35). Everything to its left is an explicitly centered angular rounding error. In particular, writing $r_n = m_n - 1/2 + \alpha_n$ gives its exact summands as (15); replacing them by $r_n + 1/4$ destroys the cancellation and is too coarse for the compact range.

The remaining discontinuities are transparent. For a fixed n , the summand is smooth until r_n crosses a half-integer. At $r_n = m + 1/2$, strict counting assigns $M(r_n) = m$; immediately on the side $r_n > m + 1/2$, the summand jumps upward by $2m + 1$. These are exactly the original proxy walls, now separated from the radial Stieltjes deficit.

There is a natural second summation problem behind (37). Put

$$e(r) := M(r)^2 - r^2, \quad q(r) := -A'(r), \quad h(t) := e(R(t)). \quad (38)$$

On every complete angular cell ($m \geq 1$),

$$\int_{m-1/2}^{m+1/2} (m^2 - r^2) dr = -\frac{1}{12}. \quad (39)$$

Thus the continuous comparison

$$\int_0^T h(t) dt = \int_0^K e(r)q(r) dr$$

has a favorable cell-average drift. Passing from this integral to the shifted samples $h(n-1/4)$ produces the genuine double-sawtooth correlation. Its atoms contain the proxy-wall jumps and cannot simply be discarded by sign. In particular, if an angular wall coincides with a radial sample, Stieltjes integration by parts must retain the common-jump term dictated by the strict convention. A successful next lemma must pair those jump terms with the positive quantity already retained in $\mathcal{H}$, or bound them in blocks using the single-peaked shape of q from (6).

Remark 5 (Status). All identities and inequalities through Corollary 4 are analytic and require no finite computation. Equation (37) is a proposed replacement lemma, not a proved claim. Consequently this note does not by itself remove the compact certificate from the spherical-shell proof.